The prime objectives of creating smart city environments are: optimised decision-making, building infrastructure, and the use of cyber and physical resources to address the challenges in urban areas. However, the large-scale deployment of cyber-physical-social systems using open source has its own set of challenges that require smarter sensing and computing methods, as well as advanced networking and communication technologies to provide better services.

Open source for next-gen networks

Gone are the days when 5G, IoT and edge computing were just buzzwords. In today's business verticals, these are quite the reality. Open source is shaping large areas of technology. For example, in telecommunications, it is not just a way to foster collaborative research and innovation, but an opportunity to make real change in the telco ecosystem. Open source projects are creating technology that will drive the evolution of the next-generation mobile networks, which is vital as industries move towards the 5G era.

"It is interesting to know that 80 per cent of telecom data is non-differentiated and can be used as open source. Instead of just consuming it, we can use the best it offers," says Dr Aloknath De, CTO, Samsung India.

The concept of collaborative development on networks and new technologies is not new in telecom. "Samsung uses the Tizen platform to build digital appliances, and Jerryscript from JavaScript is intended to run on a very constrained kind of environment. It also supports IoT and Open Connectivity Foundation," says Dr De. Moreover, when it comes to 5G communication, he considers Open Radio Access Network Software Community (O-RAN SC) and Open Networking Automation Platform (ONAP) as two big elements. Akraino is also an important element that supports high availability of cloud services, spanning a variety of use cases for artificial intelligence.

Fundamental shift in architecture

Industries are today focusing on product consumption, software architecture, modularity of software network functions, application services, and use of open source, all of which indicates a major shift in architecture.

Dr Aloknath De,
CTO, Samsung India

Gnanapriya Chidambaranathan,
AVP and senior principal architect, Infosys

Subodh Gajare, lead architect
(5G and IoT security), Cisco R&D

Subodh Gajare, lead architect (5G and IoT security), Cisco R&D, feels that nowhere between 2G and 4G, did we witness such major architectural shifts. "When we look at 5G on a silo compared to the three previous generations of mobility, we see a fundamental shift in architecture in the way network components are looked at, and the security element has been blown apart."

He adds, "We are witnessing three major architectural shifts, and this is the place where 7 trillion devices, 7 trillion people and all these economies of scale will be met. And that's why it's both a huge competitive advantage and a huge secondary challenge as well."

Innovation in the open network ecosystem

Gnanapriya Chidambaranathan, AVP and senior principal architect, Infosys, says that we are looking towards how open networking is bringing in innovation, embracing the cloud nativeness as well as dynamic orchestration and automation of network slices, bringing closed loop assurance and exposing open APIs for ecosystem integration.

For instance, all of us are connected remotely and enterprise workloads are moving from the data centres to the edge. Similarly, when we talk about the industry verticals, whether it is manufacturing or Industry 4.0, a lot of low latency analytics and insights are required. And from a consumer perspective, users can watch high definition, video live streaming and other immersive experiences.

"There are a variety of possibilities that exist today. Open source brings in that openness and helps in driving innovation and cost efficiency," she says.

RANs enable physical access to devices and were mostly developed as complete proprietary solutions. This means that it was difficult for innovations to happen at the same pace as the rest of the market.